Session-8

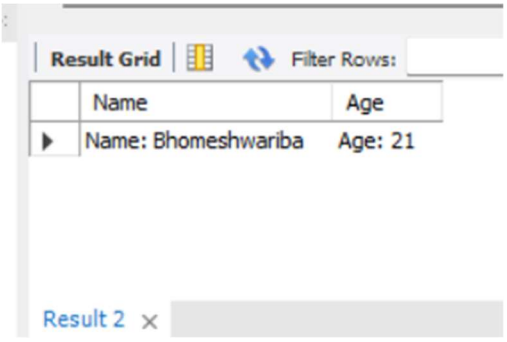
Question1: Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

Ans:

```
SET @user_name = 'Bhomeshwariba';

SET @user_age = 21;

SELECT CONCAT('Name: ', @user_name) AS Name, CONCAT('Age: ', @user_age) AS
Age;
```



Question2: Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.

Ans:

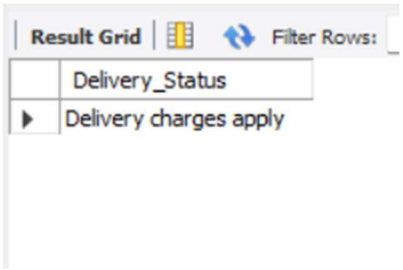
```
set @order_amount =500;

select if(@order_amount > 500,

'Eligible for free delivery',

'Delivery charges apply')

as Delivery_Status;
```



Question3: Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE.

Hint: Use a counter variable and EXIT WHEN condition inside the loop.

Ans:

```
USE counter_mydb;

DELIMITER //

CREATE PROCEDURE print_numbers()

BEGIN

   DECLARE counter INT DEFAULT 1;

 simple_loop: LOOP

      SELECT counter;

SET counter = counter + 1;

      IF counter > 5 THEN

         LEAVE simple_loop;

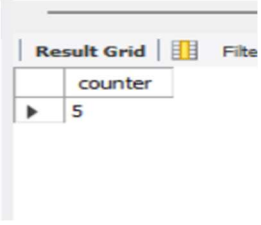
      END IF;

   END LOOP;

END //

DELIMITER ;

CALL print_numbers .
```



Question4: Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE.

Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.

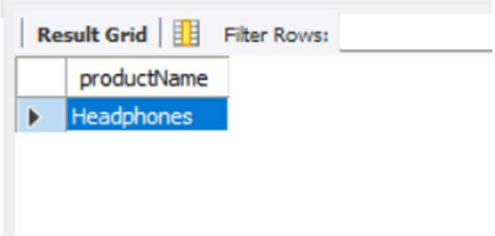
Ans:

```
USE mydb;


DROP PROCEDURE IF EXISTS show_product_names;
INSERT INTO PRODUCTS (PRODUCT_NAME) VALUES
('Laptop'),
('Mobile'),
('Keyboard'),
('Mouse'),
('Headphones');


DELIMITER //
CREATE PROCEDURE show_product_names()
BEGIN
   DECLARE done INT DEFAULT 0;
   DECLARE productName VARCHAR(255);
    DECLARE product_cursor CURSOR FOR
    SELECT PRODUCT_NAME FROM PRODUCTS;
  DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
  OPEN product_cursor;
  read_loop: LOOP
  FETCH product_cursor INTO productName;
  IF done = 1 THEN
  LEAVE read_loop;
   END IF;
 SELECT productName;
  END LOOP;
```

```
  CLOSE product_cursor;
END //
DELIMITER ;
CALL show_product_names();
```



Question5: Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

Ans:

```
USE mydb;
DROP PROCEDURE IF EXISTS divide_numbers;
DELIMITER //
CREATE PROCEDURE divide_numbers()
BEGIN
  DECLARE num1 DECIMAL(10,2) DEFAULT 10;
  DECLARE num2 DECIMAL(10,2) DEFAULT 0;
  DECLARE result DECIMAL(10,2);
  DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
  BEGIN
    SELECT 'Cannot divide by zero' AS Message;
  END;
  IF num2 = 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Cannot divide by zero';
  ELSE
```

```
    SET result = num1 / num2;
    SELECT result AS Result;
  END IF;
END //
DELIMITER ;
CALL divide_numbers();
```

